Sion Yoon

AI/Data Scientist — LLM, RAG & NLP

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Education

Michigan State University, *BS in Data Science, Minor in Mathematics* – East Lansing, MI **Sept 2021 – present**

– Engineering Dean's List. On a gap year for mandatory military service in Korea (alternative service at Tridge)

Professional Experience

Tridge, *AI/Data Scientist* – Seoul, South Korea **Jan 2025 – present**

– Built a **RAG-based food product classifier** that maps noisy, multilingual trade descriptions to Tridge's ∼11,000-node product ontology (LLM normalization, pgvector retrieval, LLM reasoning with domain rules)

– In production since Oct 2025, classifying tens of thousands of trade records weekly at **99.7% lower cost** (∼$0.002 per record) and **45% higher accuracy** (now 95%+)

– Built the AI chatbot and algorithms for **Tridge Eye** "**My Curation**", turning a question into a personalized market radar

– Built internal data systems and a Streamlit validation tool combining user and AI inputs for sales operations

Michigan State University, *Research Assistant* – East Lansing, MI **2023–24, 2026–present**

– Built an open-source Qualtrics toolkit for **multi-agent group conversation experiments**: an AI facilitator and two AI personas with real-time, lip-synced photorealistic avatars (OpenAI, Tavus), in a single JavaScript file with no server

– Designed event-gated turn-taking that syncs text with avatar speech, plus a timing-matched text-only control

– Earlier built a multi-chatbot survey platform (Django, AWS, ChatGPT API). *Adviser: Dr. Yoo Jung (Erika) Oh*

Keimyung University (RIPL), *Research Intern* – Daegu, South Korea **May 2024 – Aug 2024**

– Built a **real-time panorama stitching pipeline** merging an autonomous ship's two front cameras (C++, ROS, OpenCV)

– Funded by KRISO (Korea Research Institute of Ships & Ocean Engineering). *Adviser: Dr. Seonghun Hong*

Michigan State University (Pyeon Lab), *PM & Full-stack Developer* – East Lansing, MI **Jan 2024 – Aug 2024**

– **Led a team of 5** to rebuild pyeonlab.org, replacing the WordPress admin with a custom Django application

Facility for Rare Isotope Beams (FRIB), *Student Research Assistant* – East Lansing, MI **Aug 2023 – May 2024**

– Built emissivity and disk-thickness measurement devices; ran thermal-structural analysis of beam targets

Leadership

LIKELION MSU, *President* – East Lansing, MI **June 2023 – May 2024**

– Led a Korean coding education group; taught web development to non-majors and managed nine hackathon projects

Publications

Tool Demo: A Facilitator-Led Multi-Agent Avatar Toolkit for Group Interaction Experiments in Qualtrics *Under review*

Sion Yoon, Yoo Jung (Erika) Oh

Designing and Evaluating Multi-Chatbot Interface for Human-AI Communication: Preliminary Findings from a Persuasion Task *June 2024*

Sion Yoon, Tae Eun Kim, Yoo Jung Oh

arxiv.org/abs/2406.19648 (arXiv preprint)

Skills

Technical: Python, SQL, LLMs/RAG, LangChain, pgvector, PostgreSQL, GCP, Snowflake, AWS

Languages: Korean, English (bilingual)